

Course: 550- Computational Machine Learning

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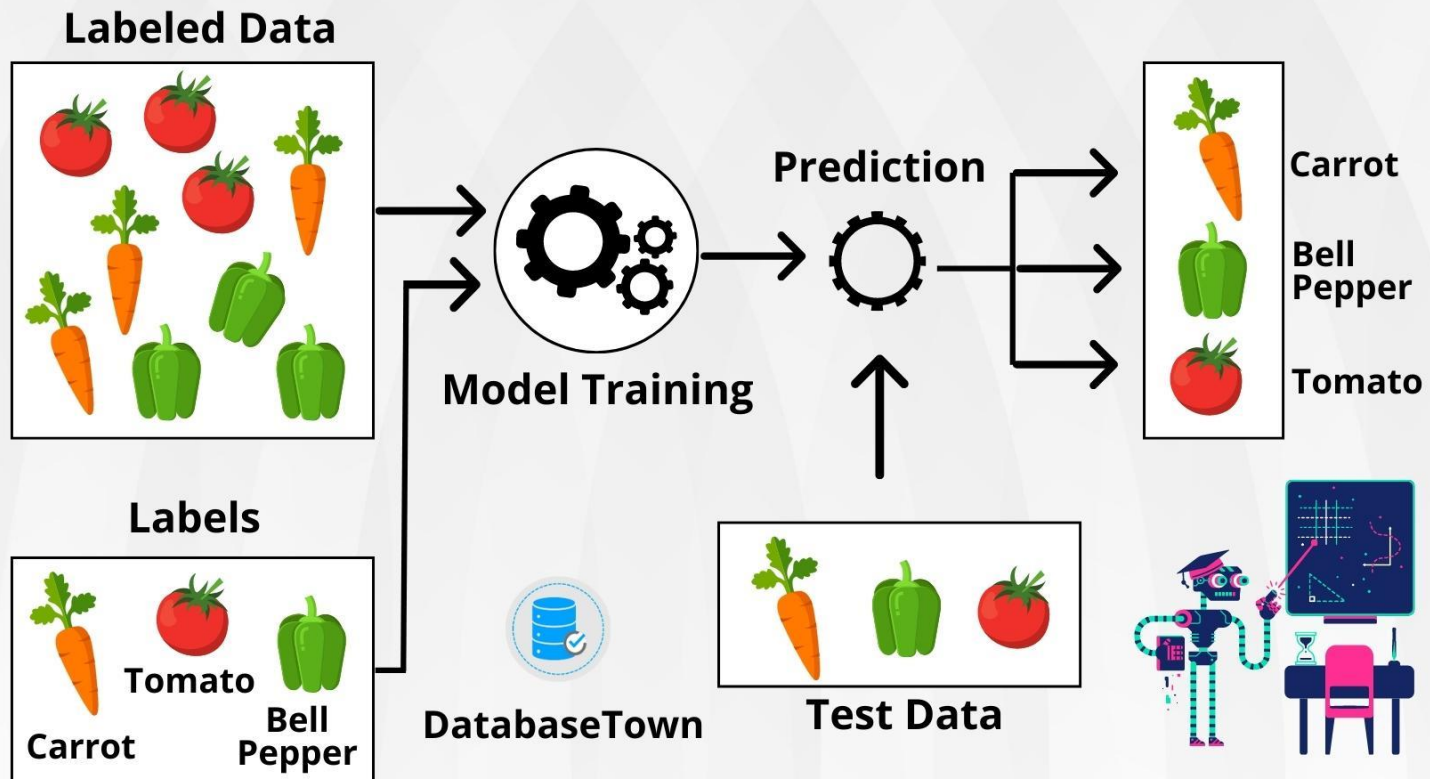
Meharry Medical College

Contents

- Brief Recap on Previous Models
- Decision Tree
- Random Forest
- Discussion on Assignment 1

SUPERVISED LEARNING

Supervised machine learning is a branch of artificial intelligence that focuses on training models to make predictions or decisions based on labeled training data.



Training Data

	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Array type
1					
2	2.25	2.83	9.13	6.66	Flex28
3	2.45	2.7	8.88	6.49	Flex28
4	3.25	2.83	9.18	6.88	Flex28
5	0.75	2.76	8.97	6.79	Standard
6	3.65	2.94	9.55	6.96	Flex28
7	3.05	2.97	9.61	7.13	Flex28
8	3.35	2.95	9.47	7.11	Flex28
9	2.35	2.8	9.04	6.57	Flex28
10	2.1	2.73	8.89	6.51	Flex28
11	2.15	2.9	9.47	6.91	Flex28
12	2.85	2.71	8.86	6.43	Flex28
13	1.8	2.78	8.9	6.71	Standard
14	3.2	3.08	9.98	7.28	Standard
15	1.7	2.58	8.45	6.2	Flex28
16	3	2.82	9.08	6.79	Flex28
17	3.65	2.85	9.25	6.76	Flex24

Training Label

Actual angular insertion depth (o)
581
637
489
637
513
449
465
545
584
454
637
661
584
600
537
461

Testing Data

	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Array type
18	-0.68	2.67	8.66	6.32	Flex28
19	3.2	2.53	8.16	6.11	Flex28
20	1.85	2.76	8.99	6.62	Flex28
21	3.25	2.75	8.86	6.56	Flex28

Testing Label

500
717
516
510



New Data

BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Array type
1.9	2.99	9.74	7.24	Flex28

Model Prediction

502

Multiple Linear Regression

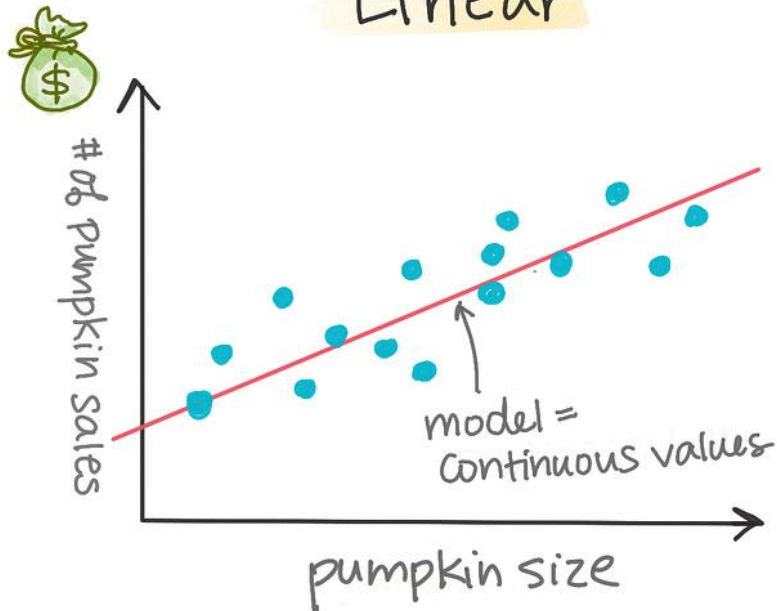
- More than one predictor...

$$E(y) = \alpha + \beta_1 * X + \beta_2 * W + \beta_3 * Z \dots$$

Each regression coefficient is the amount of change in the outcome variable that would be expected per one-unit change of the predictor, if all other variables in the model were held constant.

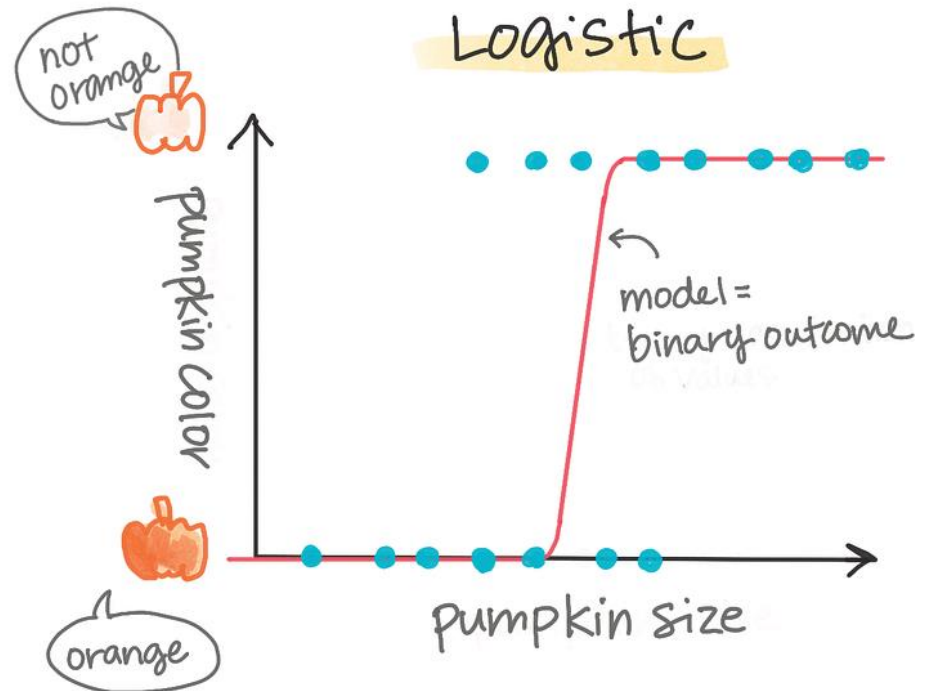
LINEAR vs. LOGISTIC REGRESSION

Linear



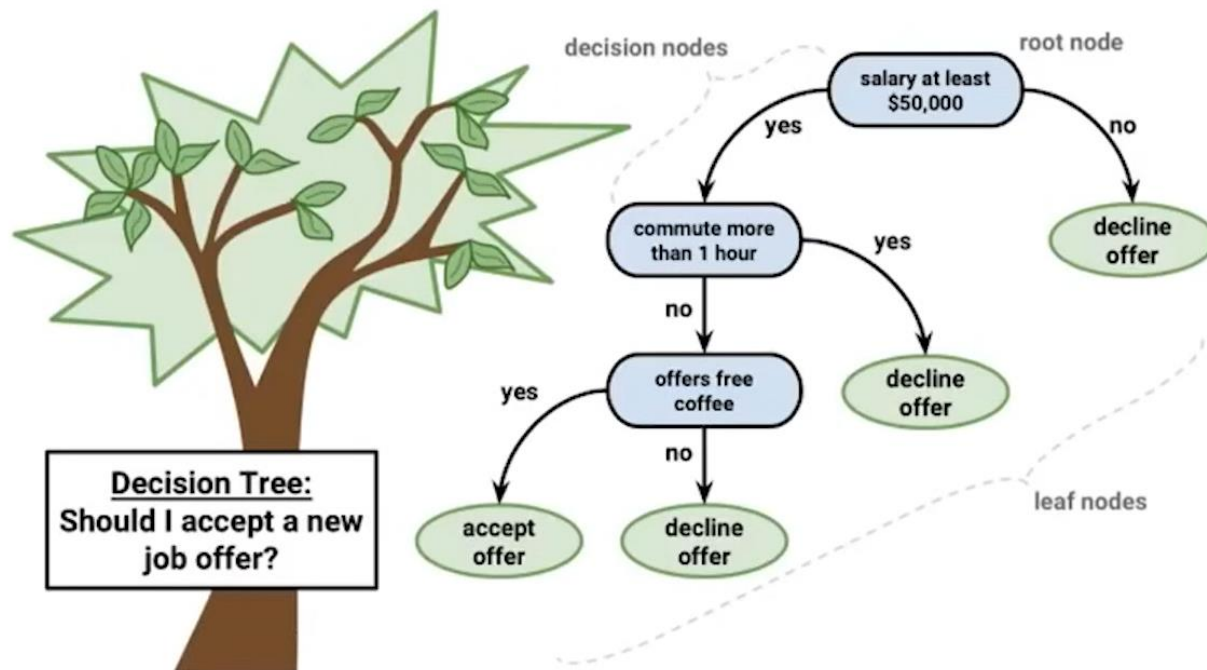
@girlie_mac

Logistic

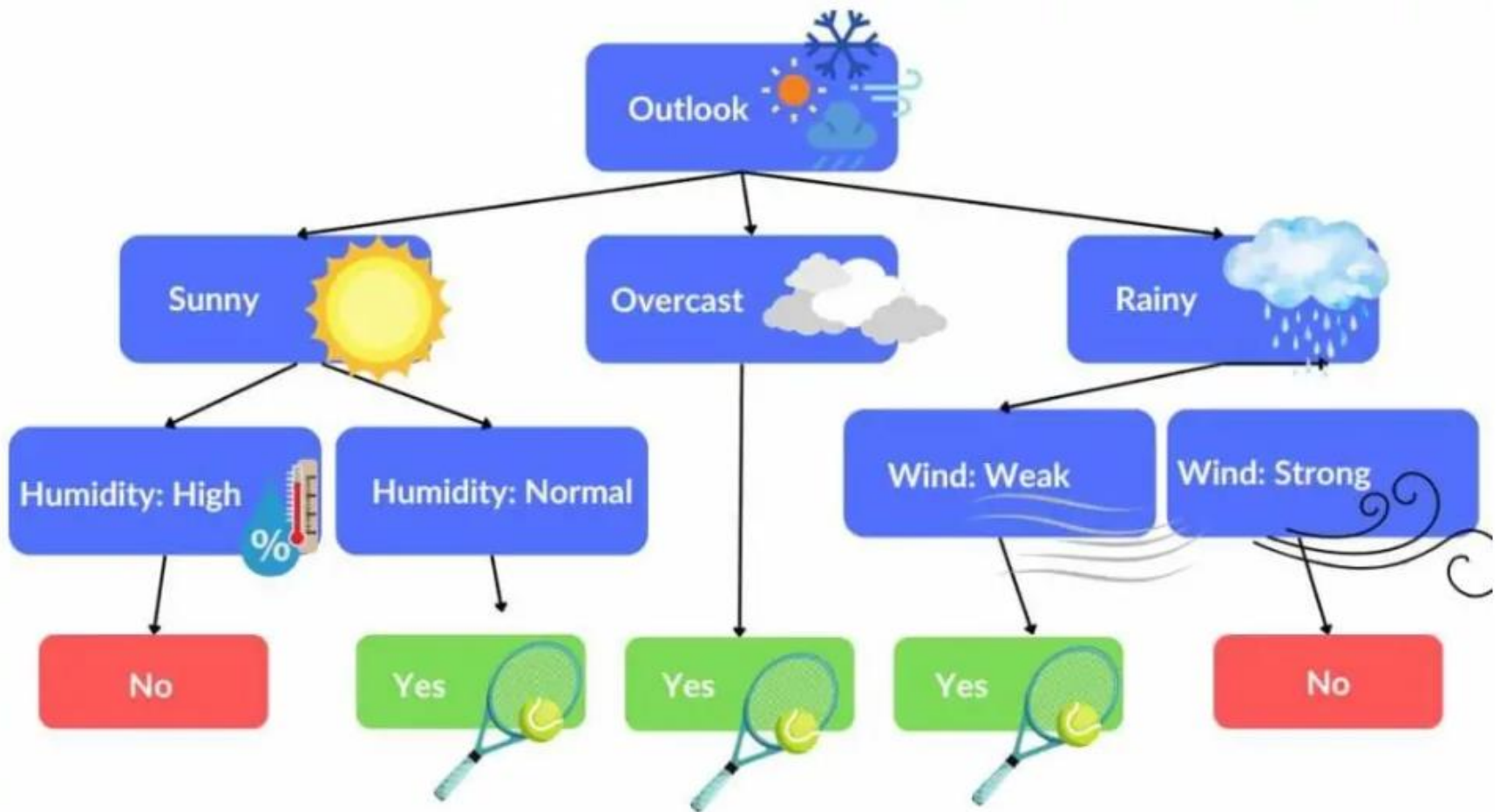


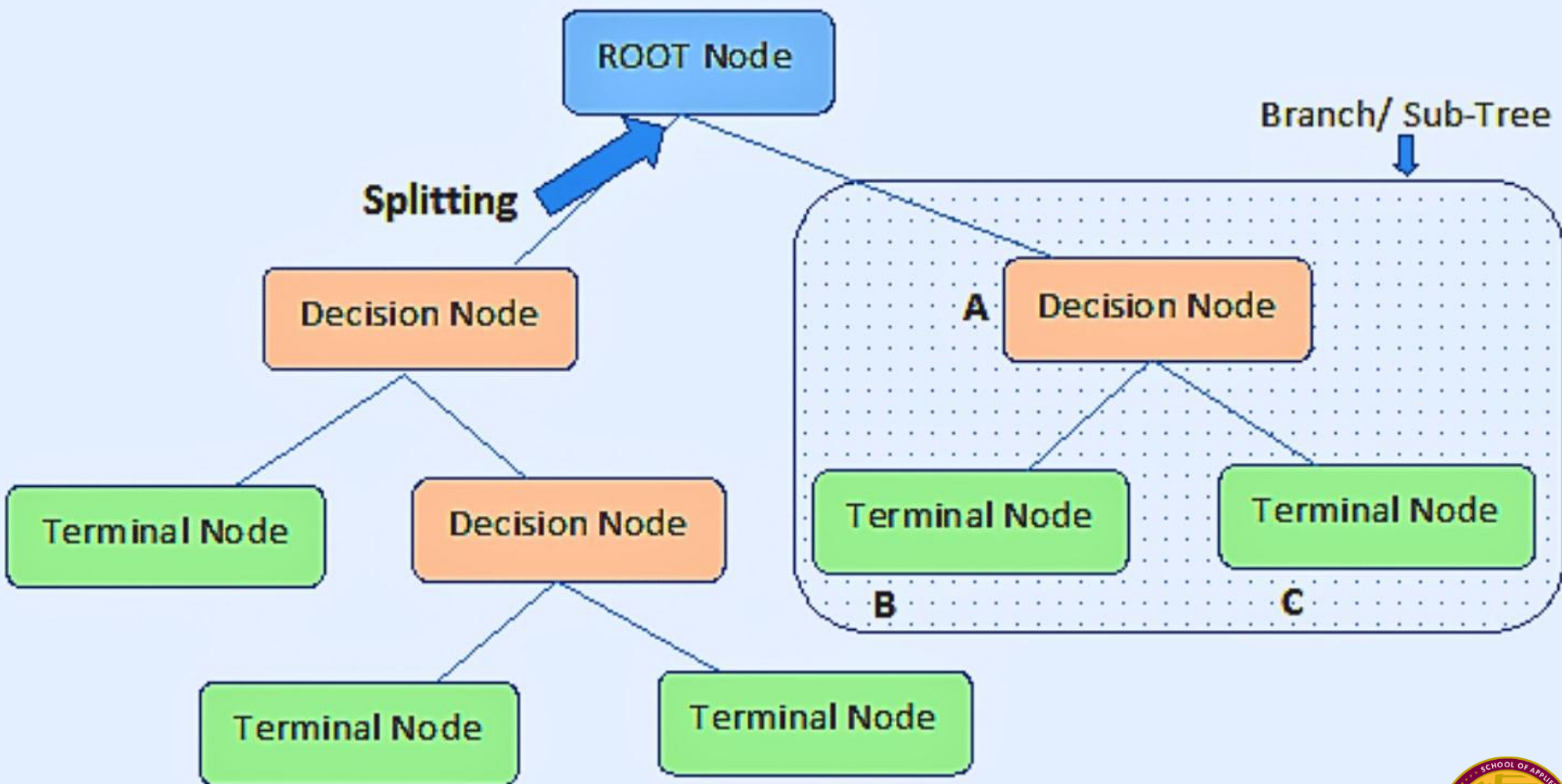
Decision Tree

“A **decision tree** is a graphical representation of all the possible solutions to a decision based on certain conditions”



Outlook	Temperature	Humidity	Wind	PlayTennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

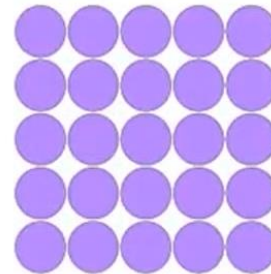




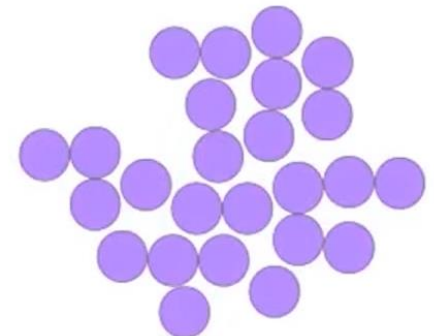
CART Algorithm

What is Entropy?

- Defines randomness in the data
- **Entropy** is just a metric which measures the impurity
- The first step to solve the problem of a decision tree

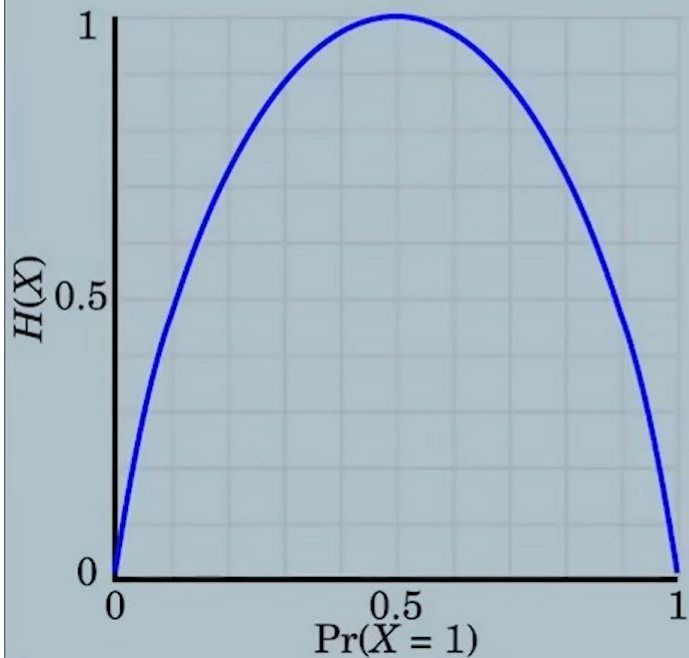


Low Entropy



High Entropy

What is Entropy?



$$\text{Entropy}(s) = -P(\text{yes}) \log_2 P(\text{yes}) - P(\text{no}) \log_2 P(\text{no})$$

Where,

- S is the total sample space,
- $P(\text{yes})$ is probability of yes

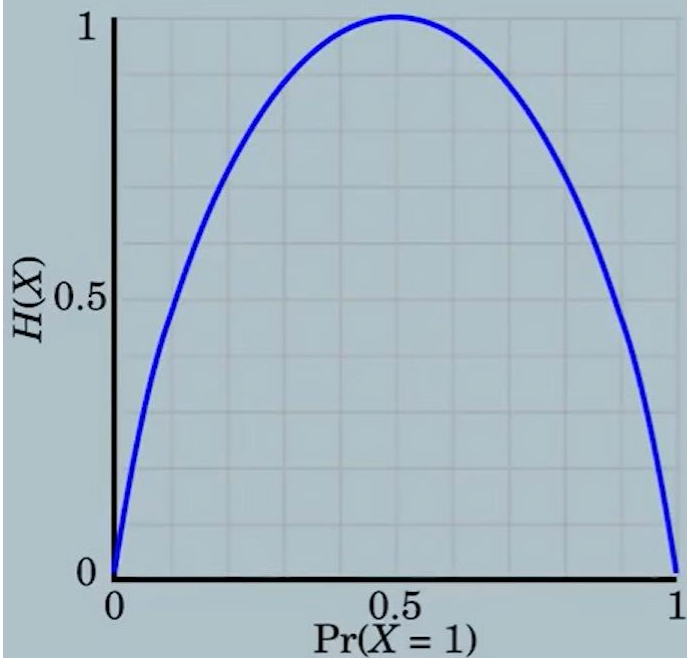
If number of **yes** = number of **no** ie $P(S) = 0.5$

$$\Rightarrow \text{Entropy}(s) = 1$$

If it contains all **yes** or all **no** ie $P(S) = 1$ or 0

$$\Rightarrow \text{Entropy}(s) = 0$$

What is Entropy?



$$E(S) = -P(\text{Yes}) \log_2 P(\text{Yes})$$

When $P(\text{Yes}) = 1$ ie YES = Total Sample(S)

$$E(S) = 1 \log_2 1$$

$$E(S) = 0$$

$$E(S) = -P(\text{No}) \log_2 P(\text{No})$$

When $P(\text{No}) = 1$ ie No = Total Sample(S)

$$E(S) = 1 \log_2 1$$

$$E(S) = 0$$

What is Information Gain?

- Measures the reduction in entropy
- Decides which attribute should be selected as the decision node

If S is our total collection,

Information Gain = Entropy(S) – [(Weighted Avg) x Entropy(each feature)]

Outlook	Temperature	Humidity	Wind	PlayTennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

Which Question to ask and When?



Which Node To Select As Root Node?

Outlook?

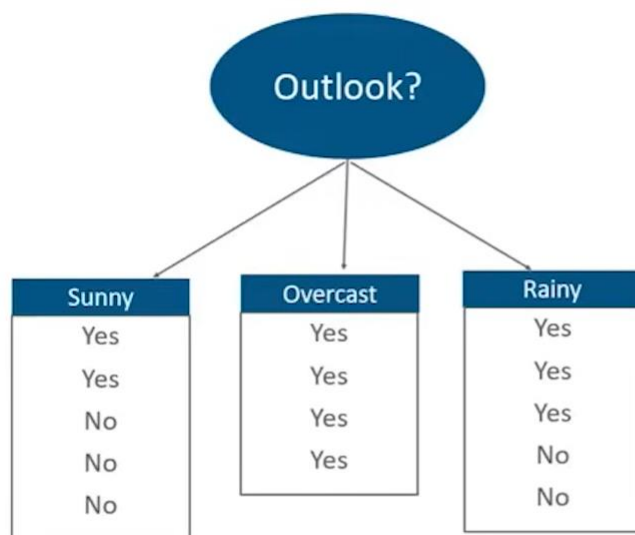
Temperature?

Humidity?

Windy?

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Which Node To Select As Root Node: Outlook



outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Which Node To Select As Root Node: Outlook

$$E(\text{Outlook} = \text{Sunny}) = -2/5 \log_2 2/5 - 3/5 \log_2 3/5 = 0.971$$

$$E(\text{Outlook} = \text{Overcast}) = -1 \log_2 1 - 0 \log_2 0 = 0$$

$$E(\text{Outlook} = \text{Rainy}) = -3/5 \log_2 3/5 - 2/5 \log_2 2/5 = 0.971$$

Information from outlook,

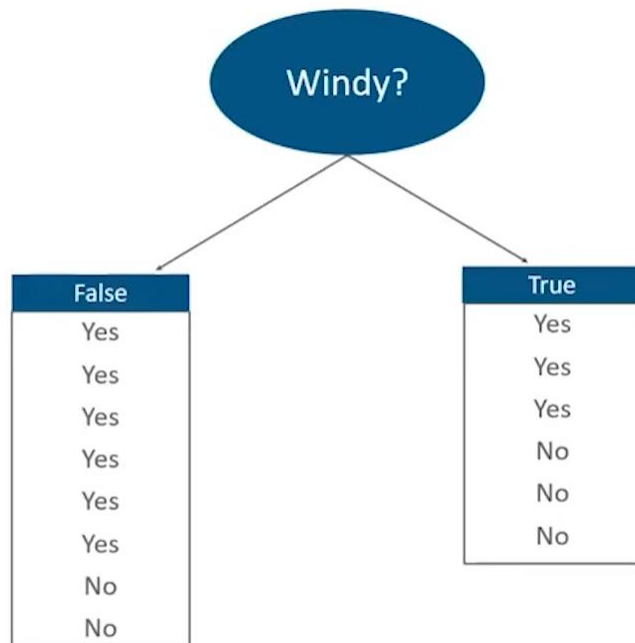
$$I(\text{Outlook}) = 5/14 \times 0.971 + 4/14 \times 0 + 5/14 \times 0.971 = 0.693$$

Information gained from outlook,

$$\text{Gain}(\text{Outlook}) = E(S) - I(\text{Outlook})$$

$$0.94 - 0.693 = 0.247$$

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no



outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Which Node To Select As Root Node: Windy

$$E(\text{Windy} = \text{True}) = 1$$

$$E(\text{Windy} = \text{False}) = 0.811$$

Information from windy,

$$I(\text{Windy}) = 8/14 \times 0.811 + 6/14 \times 1 = 0.892$$

Information gained from outlook,

$$\text{Gain}(\text{Windy}) = E(S) - I(\text{Windy})$$

$$0.94 - 0.892 = 0.048$$

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Which Node To Select As Root Node

				outlook	temp.	humidity	windy	play
Outlook: Info Gain: 0.940-0.693	0.693 0.247	Temperature: Info Gain: 0.940-0.911	0.911 0.029	sunny	hot	high	false	no
				sunny	hot	high	true	no
				overcast	hot	high	false	yes
				rainy	mild	high	false	yes
				rainy	cool	normal	false	yes
				rainy	cool	normal	true	no
				overcast	cool	normal	true	yes
Humidity: Info Gain: 0.940-0.788	0.788 0.152	Windy: Info Gain: 0.940-0.982	0.892 0.048	sunny	mild	high	false	no
				sunny	cool	normal	false	yes
				rainy	mild	normal	false	yes
				sunny	mild	normal	true	yes
				overcast	mild	high	true	yes
				overcast	hot	normal	false	yes
				rainy	mild	high	true	no

Which Node To Select As Root Node

Outlook:

Info 0.693
 Gain: 0.940-0.693

0.693
0.247

Temperature:

Info 0.911
 Gain: 0.940-0.911 0.029

Humidity:

Info 0.788
 Gain: 0.940-0.788 0.152

Windy:

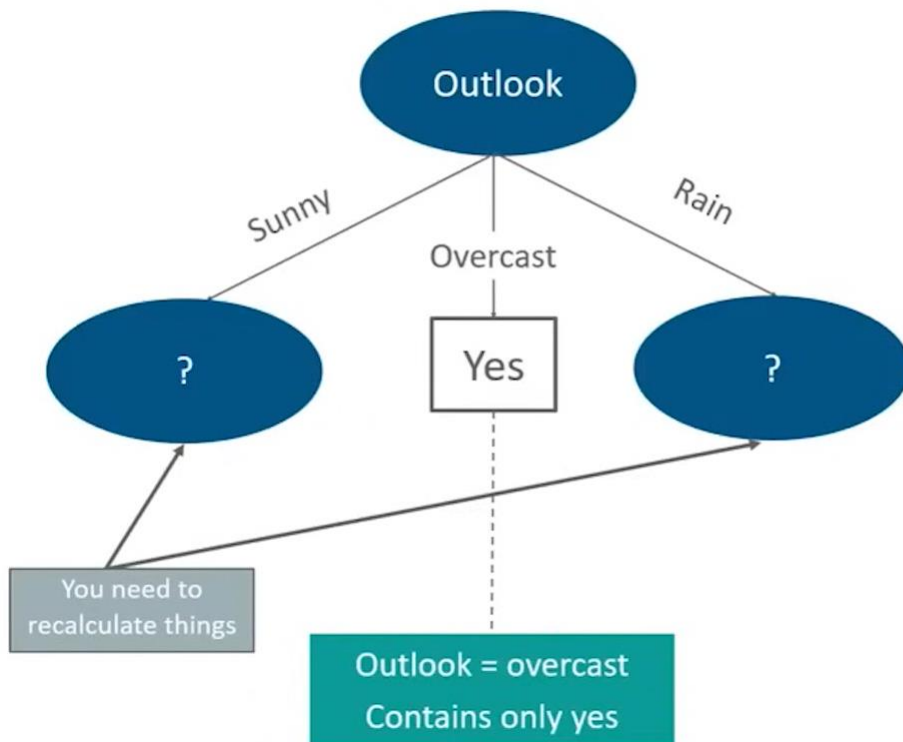
Info 0.892
 Gain: 0.940-0.982 0.048

Since Max gain = 0.247,

Outlook is our ROOT Node

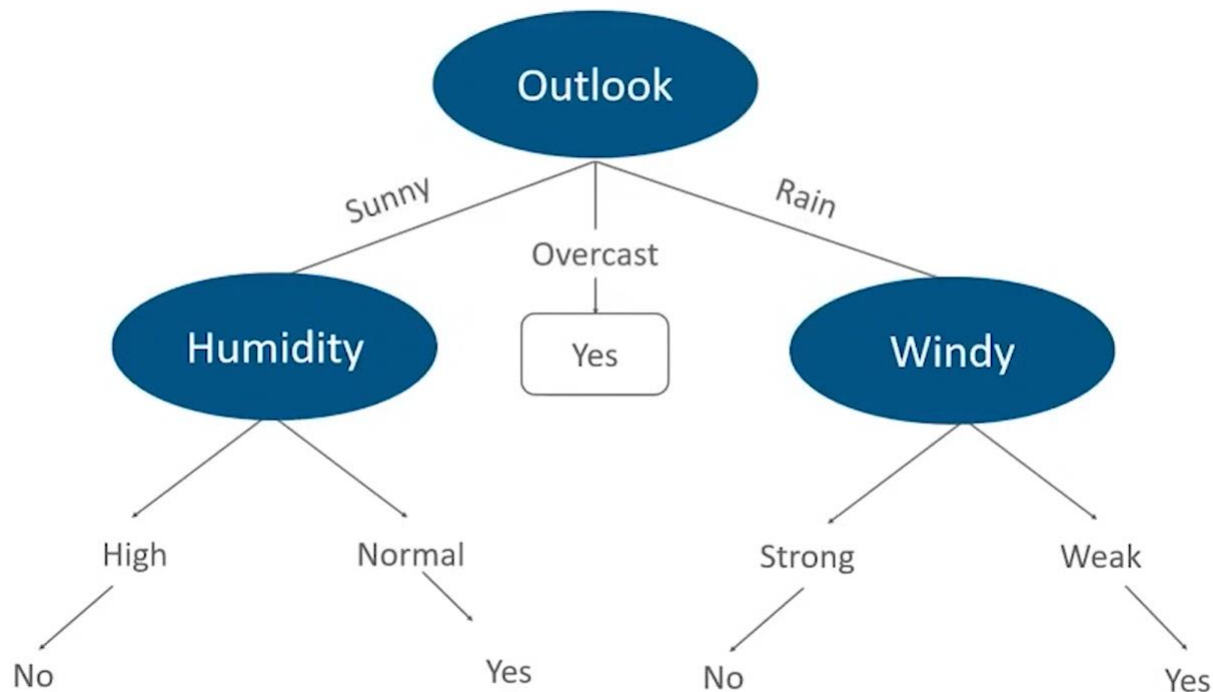
outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Which Node To Select Further?



outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

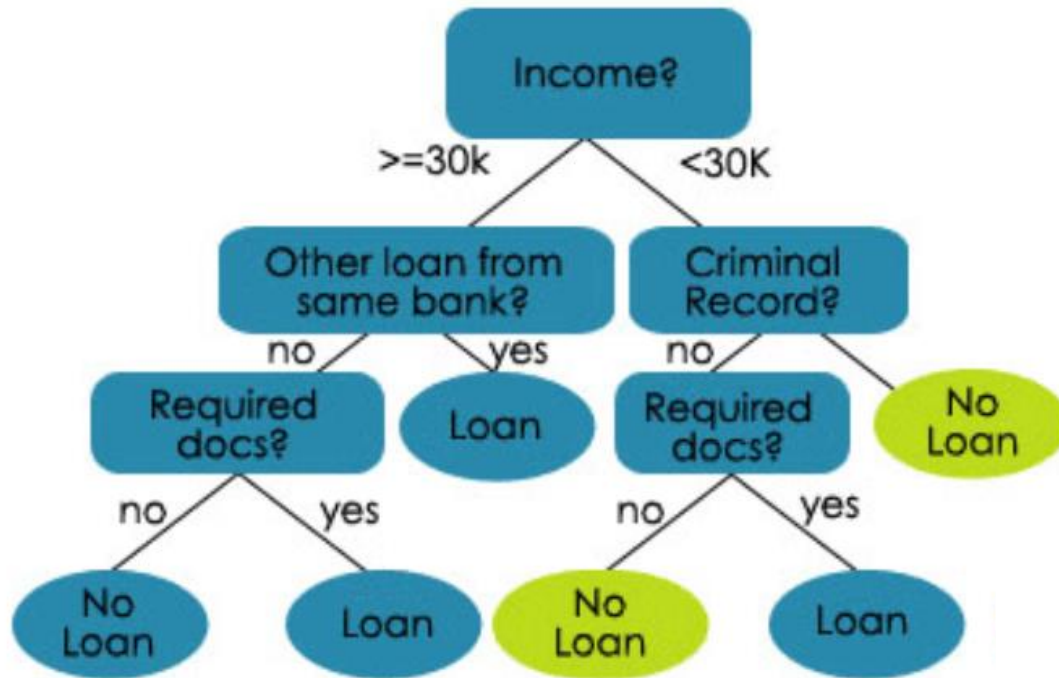
This Is How Your Complete Tree Will Look Like



Decision Tree Pruning

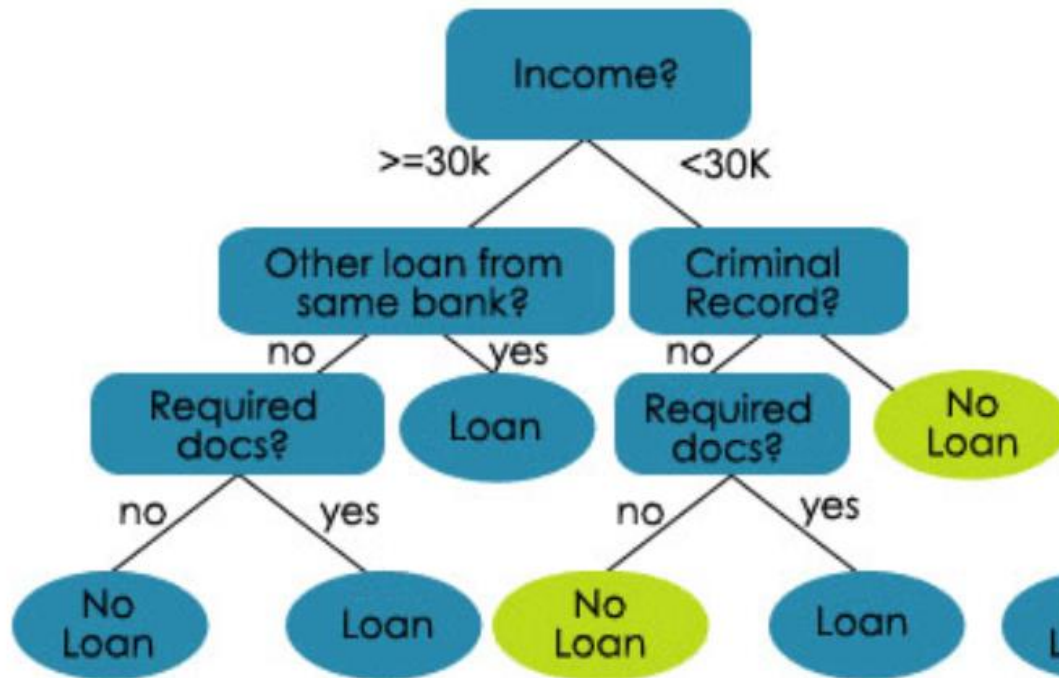
Decision Tree Pruning

An Unpruned Decision Tree

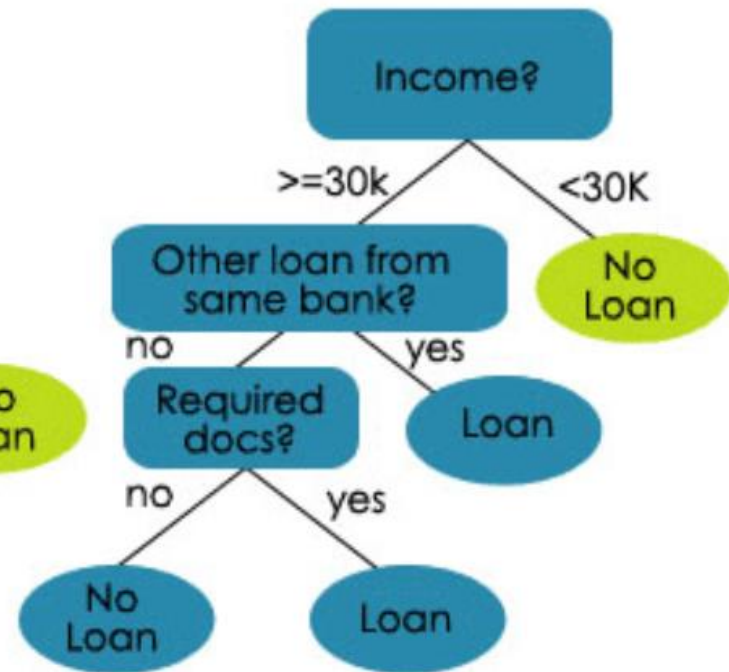


Decision Tree Pruning

An Unpruned Decision Tree



A Pruned Decision Tree



Decision Tree Pruning

Index	Age	Income	Student	Credit Rating	Buy
0	Young	High	No	Fair	No
1	Young	High	No	Excellent	No
2	Middle-aged	High	No	Fair	Yes
3	Senior	Medium	No	Fair	Yes
4	Senior	Low	Yes	Fair	Yes
5	Senior	Low	Yes	Excellent	No
6	Middle-aged	Low	Yes	Excellent	Yes
7	Young	Medium	No	Fair	No
8	Young	Low	Yes	Fair	Yes
9	Senior	Medium	Yes	Fair	Yes
10	Young	Medium	Yes	Excellent	Yes
11	Middle-aged	Medium	No	Excellent	Yes
12	Middle-aged	High	Yes	Fair	Yes
13	Senior	Medium	No	Excellent	No

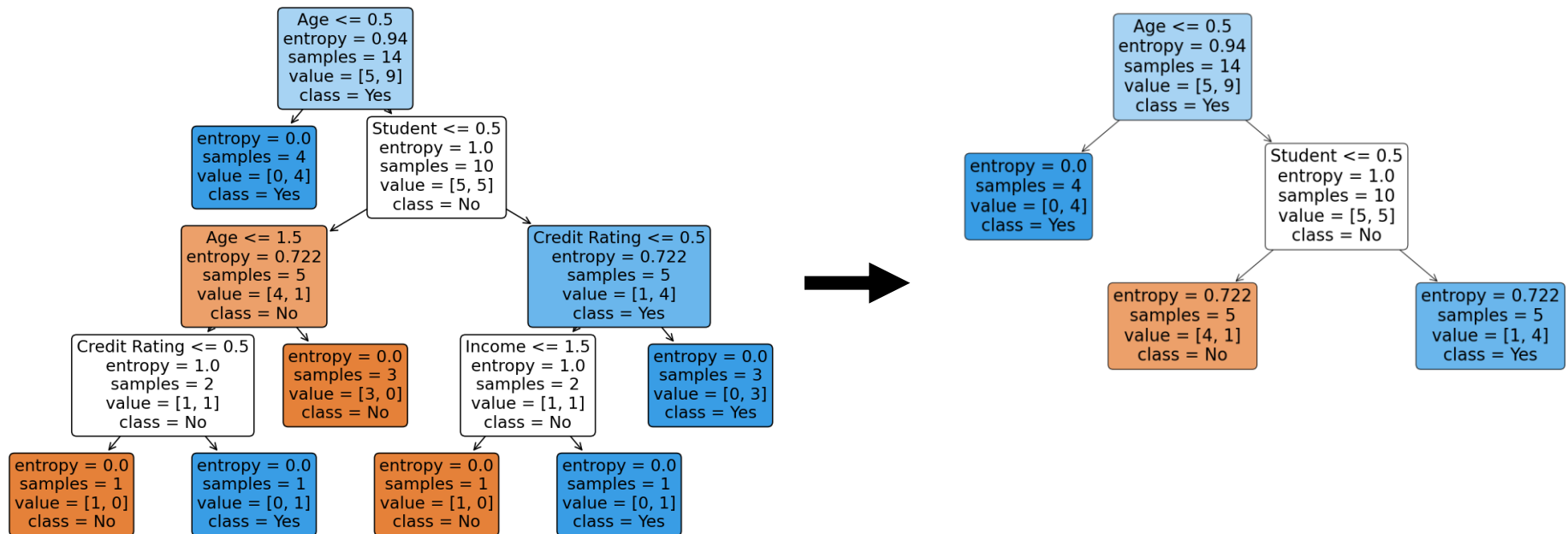
Index	Age	Income	Student	Credit Rating	Buy
0	Young	High	No	Fair	No
1	Young	High	No	Excellent	No
2	Middle-aged	High	No	Fair	Yes
3	Senior	Medium	No	Fair	Yes
4	Senior	Low	Yes	Fair	Yes
5	Senior	Low	Yes	Excellent	No
6	Middle-aged	Low	Yes	Excellent	Yes
7	Young	Medium	No	Fair	No
8	Young	Low	Yes	Fair	Yes
9	Senior	Medium	Yes	Fair	Yes
10	Young	Medium	Yes	Excellent	Yes
11	Middle-aged	Medium	No	Excellent	Yes
12	Middle-aged	High	Yes	Fair	Yes
13	Senior	Medium	No	Excellent	No



Encoding

Index	Age	Income	Student	Credit Rating	Buy
0	2	0	0	1	0
1	2	0	0	0	0
2	0	0	0	1	1
3	1	2	0	1	1
4	1	1	1	1	1
5	1	1	1	0	0
6	0	1	1	0	1
7	2	2	0	1	0
8	2	1	1	1	1
9	1	2	1	1	1
10	2	2	1	0	1
11	0	2	0	0	1
12	0	0	1	1	1
13	1	2	0	0	0

Decision Tree Pruning



Decision Tree Demo

Index	Age	Income	Student	Credit Rating	Buy
0	Young	High	No	Fair	No
1	Young	High	No	Excellent	No
2	Middle-aged	High	No	Fair	Yes
3	Senior	Medium	No	Fair	Yes
4	Senior	Low	Yes	Fair	Yes
5	Senior	Low	Yes	Excellent	No
6	Middle-aged	Low	Yes	Excellent	Yes
7	Young	Medium	No	Fair	No
8	Young	Low	Yes	Fair	Yes
9	Senior	Medium	Yes	Fair	Yes
10	Young	Medium	Yes	Excellent	Yes
11	Middle-aged	Medium	No	Excellent	Yes
12	Middle-aged	High	Yes	Fair	Yes
13	Senior	Medium	No	Excellent	No

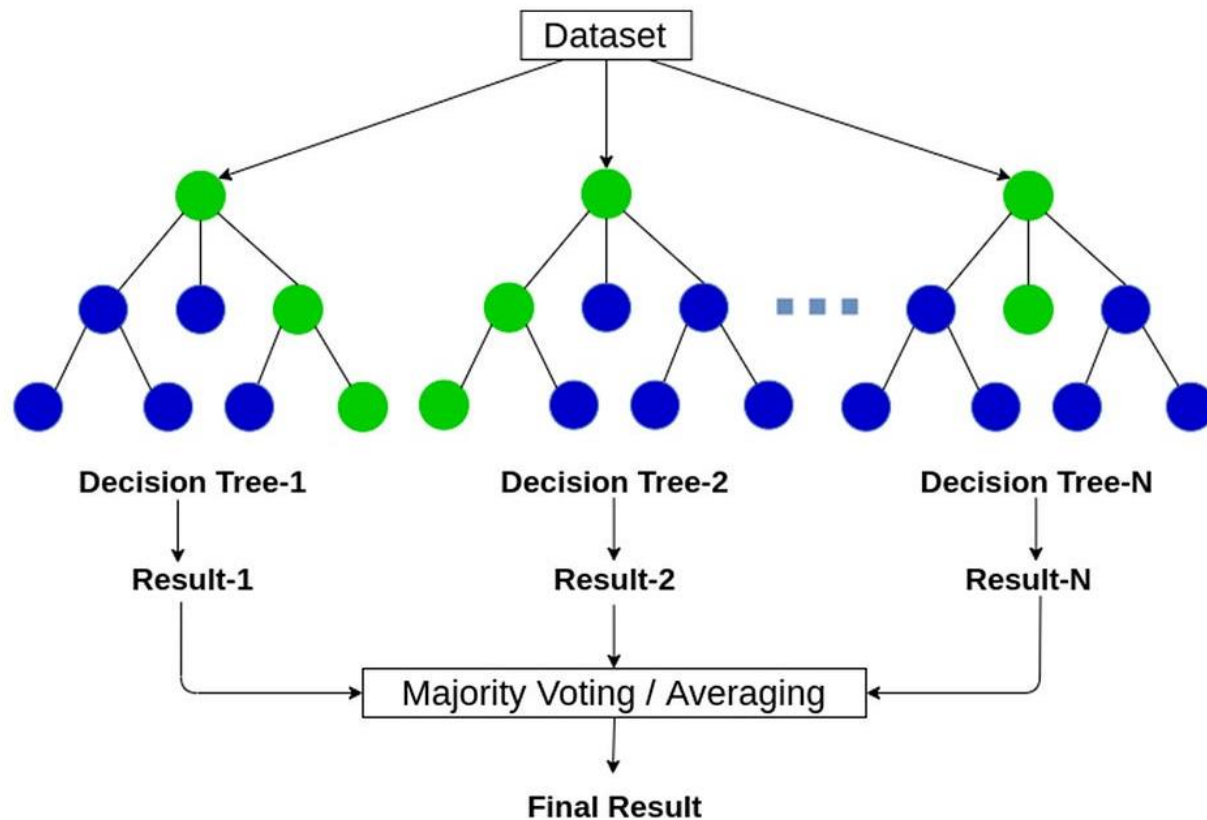
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2	Middle-aged	High	No	Fair	Yes
3	Senior	Medium	No	Fair	Yes
4	Senior	Low	Yes	Fair	Yes
5	Senior	Low	Yes	Excellent	No
6	Middle-aged	Low	Yes	Excellent	Yes
7	Young	Medium	No	Fair	No
8	Young	Low	Yes	Fair	Yes
9	Senior	Medium	Yes	Fair	Yes
10	Young	Medium	Yes	Excellent	Yes
11	Middle-aged	Medium	No	Excellent	Yes
12	Middle-aged	High	Yes	Fair	Yes
13	Senior	Medium	No	Excellent	No

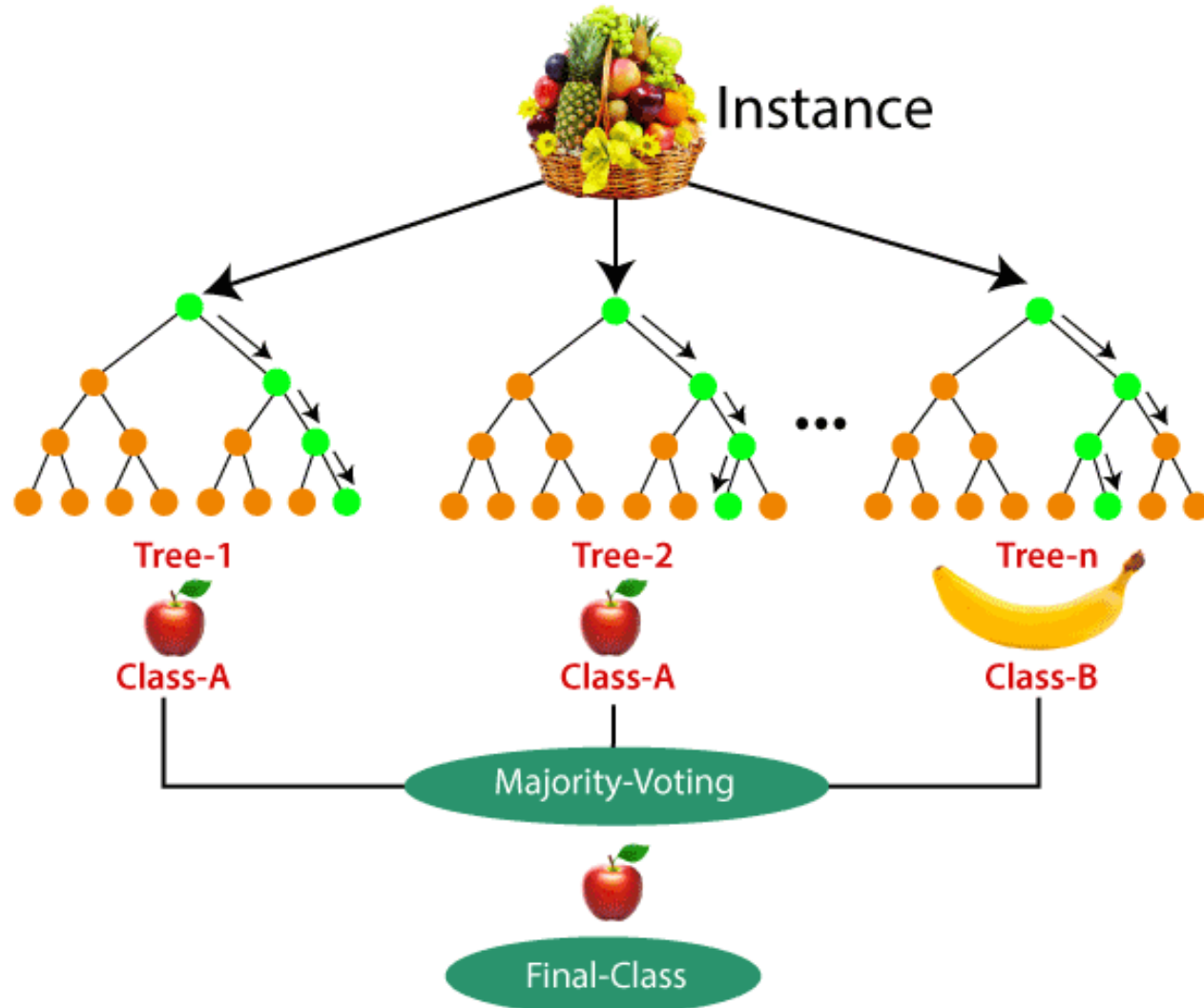


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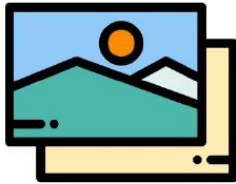
Index	Age	Income	Student	Credit Rating	Buy
0	2	0	0	1	0
1	2	0	0	0	0
2	0	0	0	1	1
3	1	2	0	1	1
4	1	1	1	1	1
5	1	1	1	0	0
6	0	1	1	0	1
7	2	2	0	1	0
8	2	1	1	1	1
9	1	2	1	1	1
10	2	2	1	0	1
11	0	2	0	0	1
12	0	0	1	1	1
13	1	2	0	0	0

Random Forest





Application of Random Forest



Remote
Sensing

Used in ETM devices
to acquire images of
the earth's surface.

Accuracy is higher
and training time is
less



Object Detection

Multiclass object
detection is done
using Random
Forest algorithms

Provides better
detection in
complicated
environments



Kinect

Random Forest
is used in a
game console
called Kinect

Tracks body
movements and
recreates it in
the game

Application of Random Forest



User performs a step



Kinect registers the movement



Marks the user based on accuracy

Why Random Forest?



No overfitting

Use of multiple trees
reduce the risk of
overfitting

Training time is less



High accuracy

Runs efficiently on
large database

For large data, it
produces highly
accurate
predictions



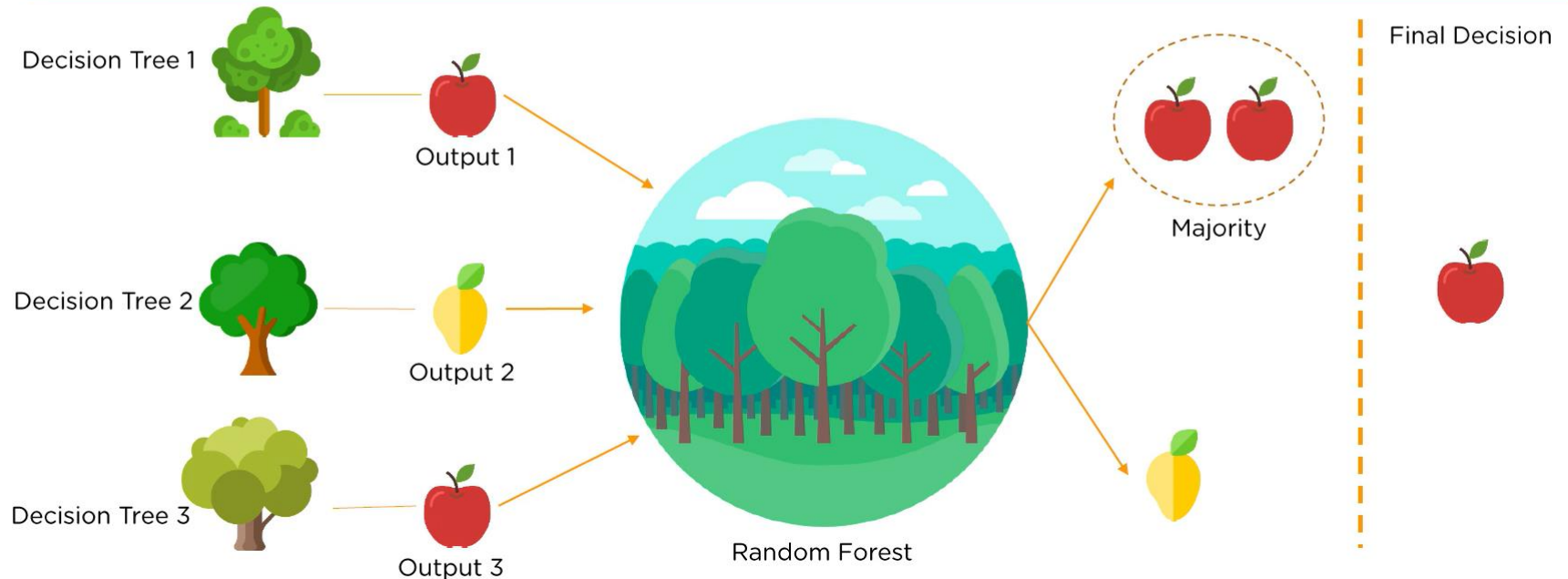
Estimates missing data

Random Forest
can maintain
accuracy when a
large proportion
of data is
missing

What is Random Forest?

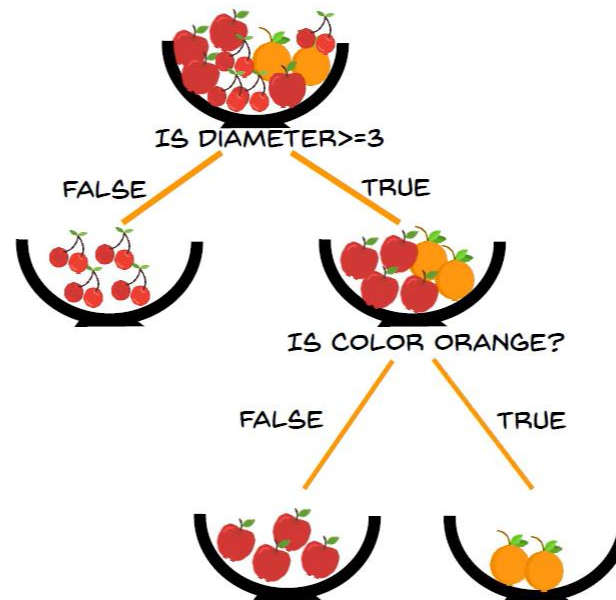
Random forest or Random Decision Forest is a method that operates by constructing multiple Decision Trees during training phase.

The Decision of the majority of the trees is chosen by the random forest as the final decision



Decision Tree

Decision Tree is a tree shaped diagram used to determine a course of action. Each branch of the tree represents a possible decision, occurrence or reaction



Decision Tree - Important Terms

Entropy

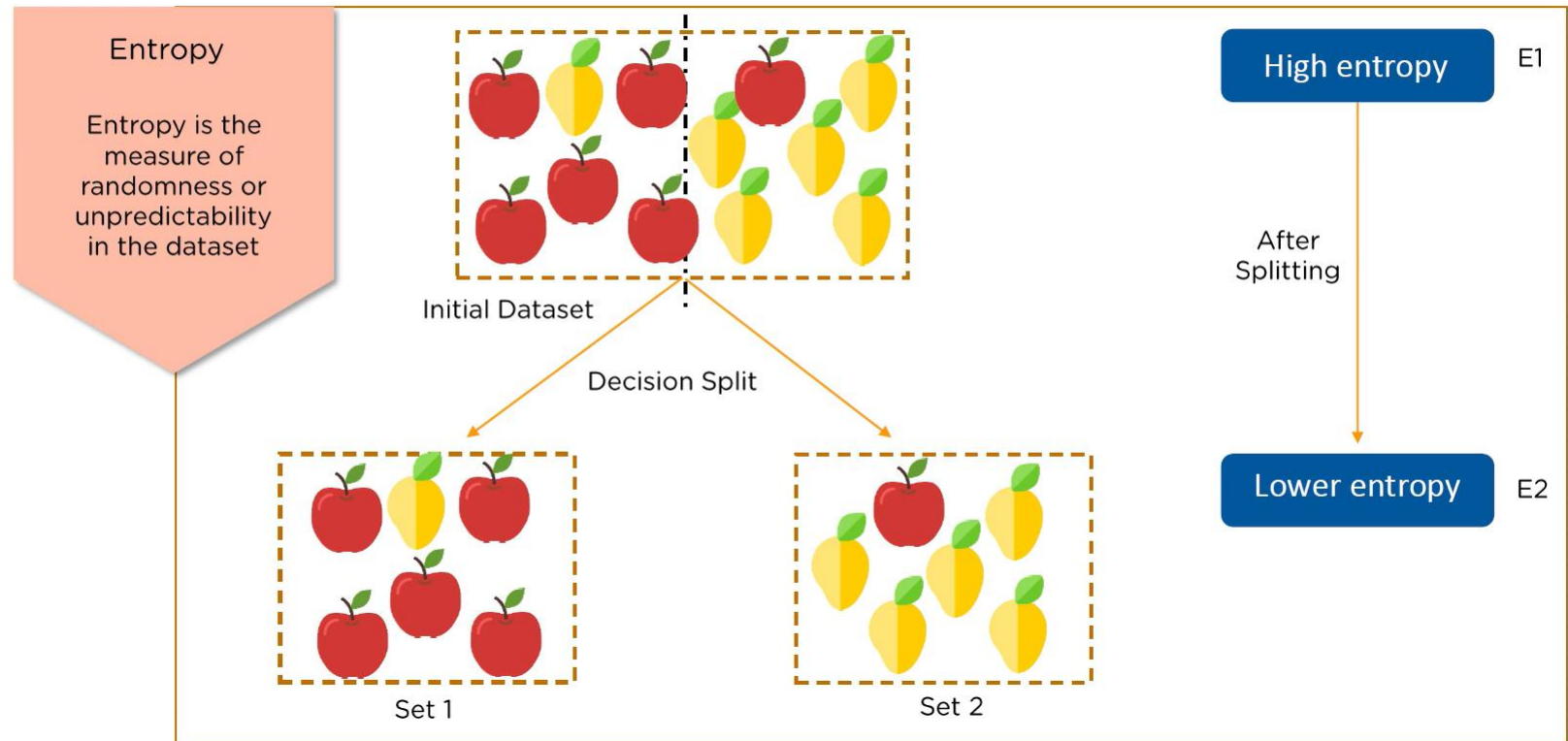
Entropy is the measure of randomness or unpredictability in the dataset



High entropy

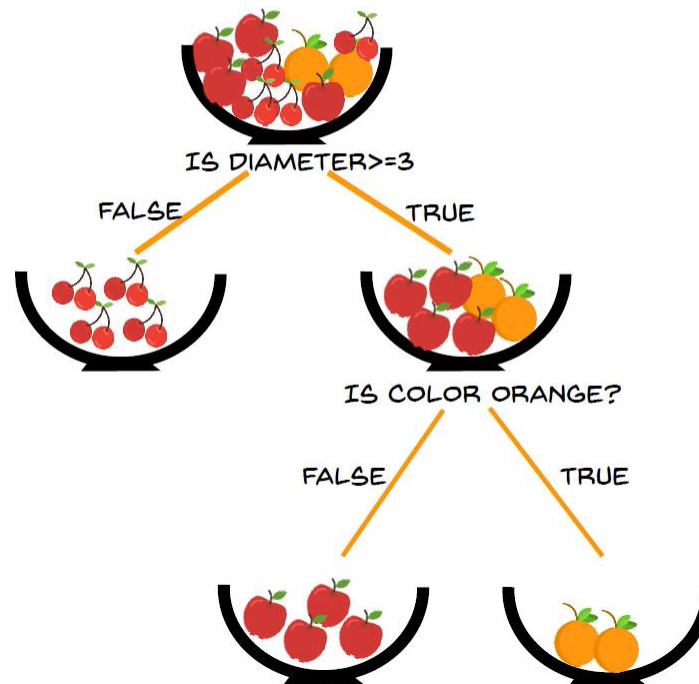
E1

Decision Tree - Important Terms



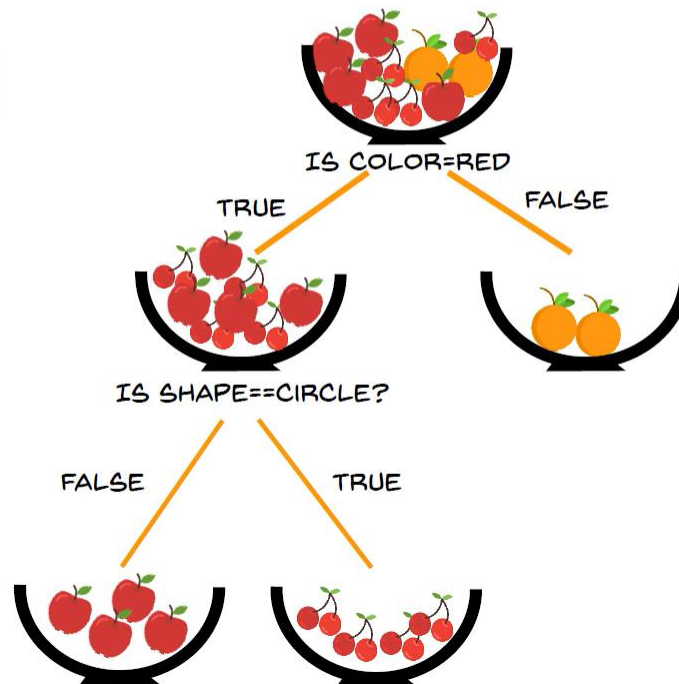
How does a Random Forest work?

LET THIS BE TREE 1



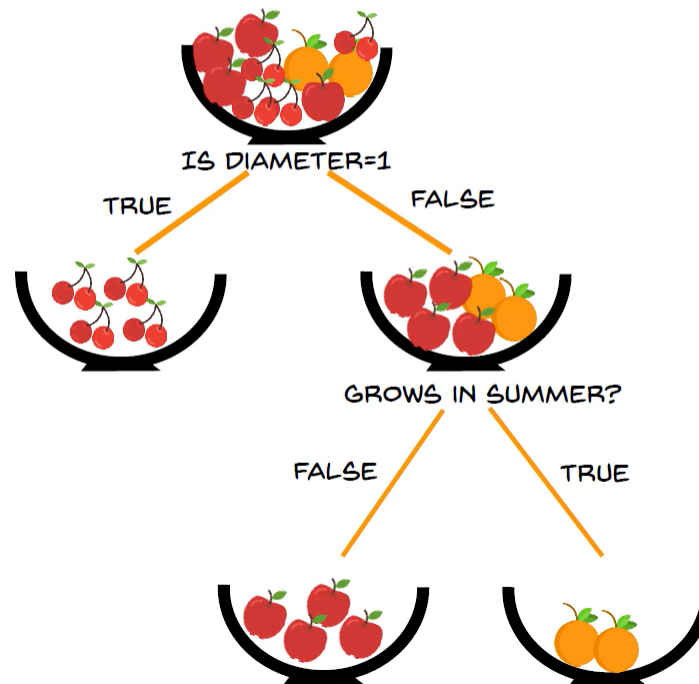
How does a Random Forest work?

LET THIS BE TREE 2

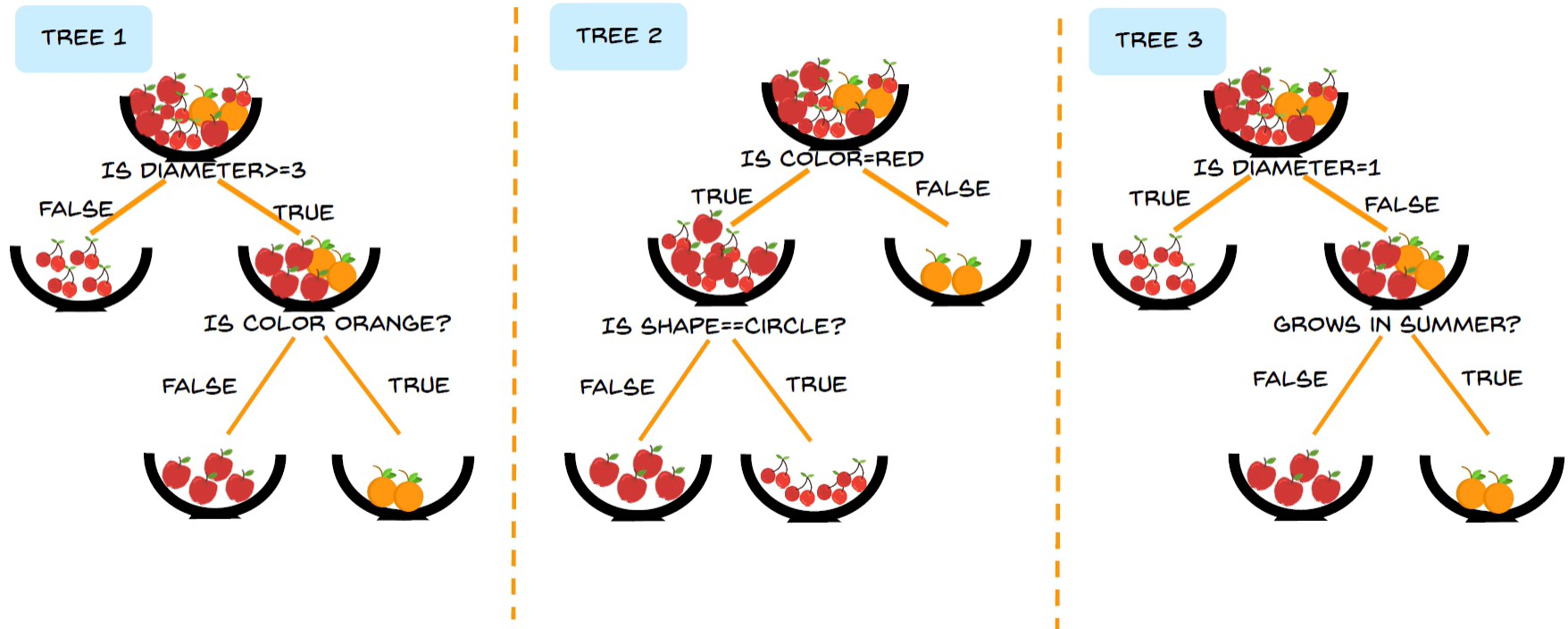


How does a Random Forest work?

LET THIS BE TREE 3



How does a Random Forest work?



How does a Random Forest work?

NOW LETS TRY TO
CLASSIFY THIS FRUIT

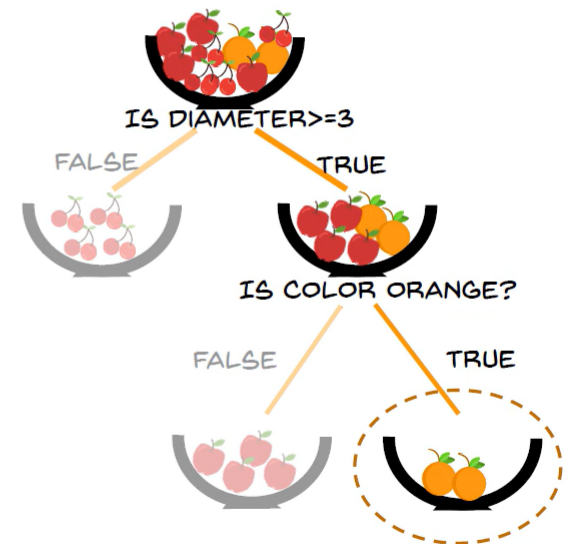


How does a Random Forest work?

TREE 1 CLASSIFIES
IT AS AN ORANGE



DIAMETER = 3
~~COLOUR = ORANGE~~
GROWS IN SUMMER = YES
SHAPE = CIRCLE

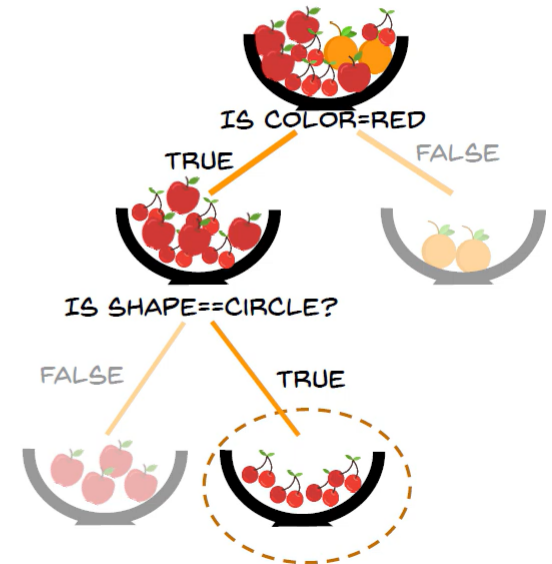


How does a Random Forest work?

TREE 2 CLASSIFIES
IT AS CHERRIES



DIAMETER = 3
~~COLOUR = ORANGE~~
GROWS IN SUMMER = YES
SHAPE = CIRCLE

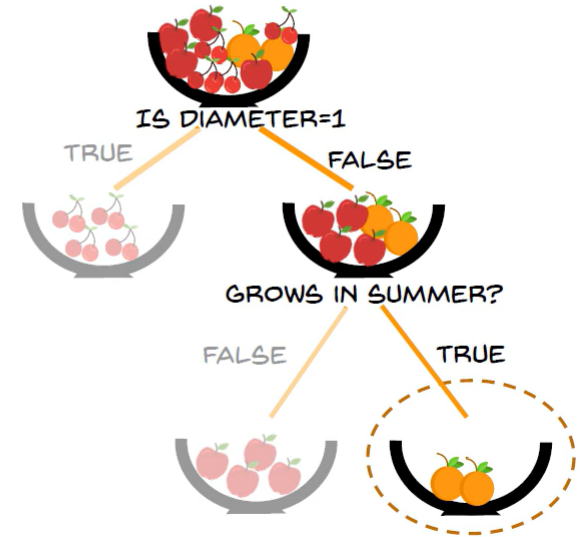


How does a Random Forest work?

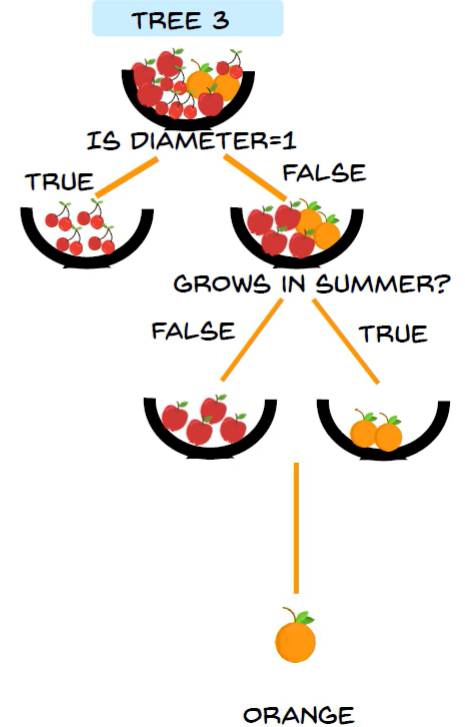
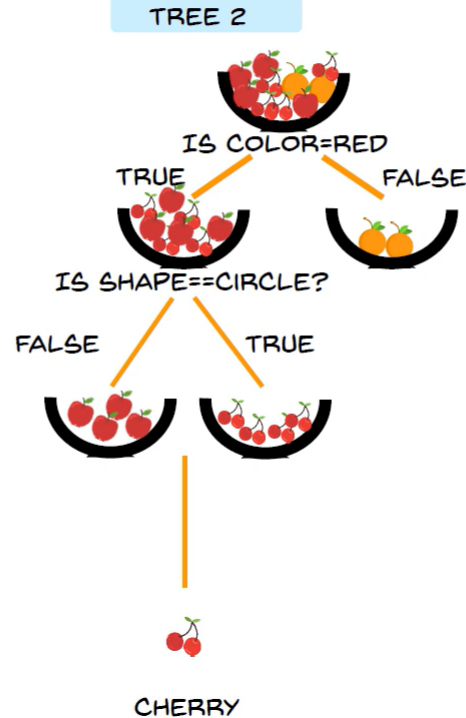
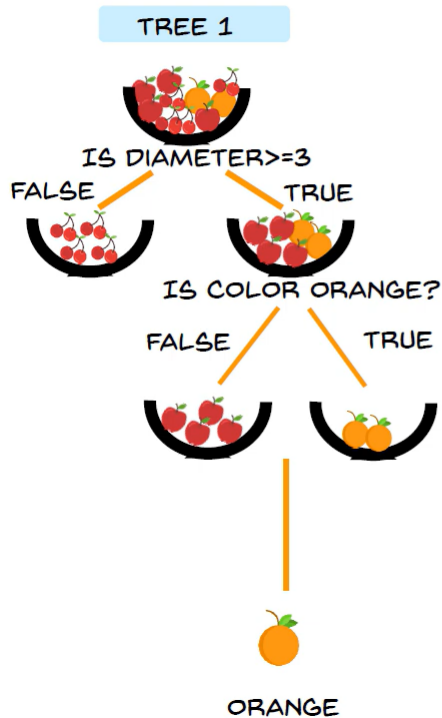
TREE 3 CLASSIFIES
IT AS ORANGE



DIAMETER = 3
~~COLOUR = ORANGE~~
GROWS IN SUMMER = YES
SHAPE = CIRCLE



How does a Random Forest work?



How does a Random Forest work?



Standardization

Standardization (also known as **feature scaling**) is a crucial step in the preprocessing of data for many machine learning algorithms, and it ensures that all features are on a similar scale. Here's why standardization is important:

1. Consistency Across Features

- Many machine learning algorithms (like **K-Nearest Neighbors**, **Support Vector Machines**, and **Gradient Descent-based methods**) rely on distance-based measures or gradient optimization, which can be heavily affected by the scale of the features.
- If one feature has a very large scale (e.g., **age** in years: 20-80) and another has a small scale (e.g., **BMI**: 10-40), the algorithm might give more importance to the feature with the larger scale, even though both features might be equally important in predicting the outcome.

2. Distance-based Algorithms

- For algorithms like **K-Nearest Neighbors (KNN)**, **K-Means Clustering**, or **Support Vector Machines (SVM)**, where distances between data points are calculated, it's important that the features contribute equally to the distance calculation.
- Without standardization, the feature with the larger range will dominate the distance calculation, causing the algorithm to perform poorly.

3. Regularization

- **Regularization techniques** (e.g., **Lasso** and **Ridge regression**) add a penalty to the coefficients of the model to avoid overfitting.



How Standardization Works:

The standardization process typically transforms each feature to have:

- **Mean = 0** (i.e., centered around the origin)
- **Standard deviation = 1** (i.e., spread out equally)

The formula for standardizing a feature is:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- **x** is the original value of the feature,
- **μ** is the mean of the feature,
- **σ** is the standard deviation of the feature,
- **z** is the standardized value of the feature.



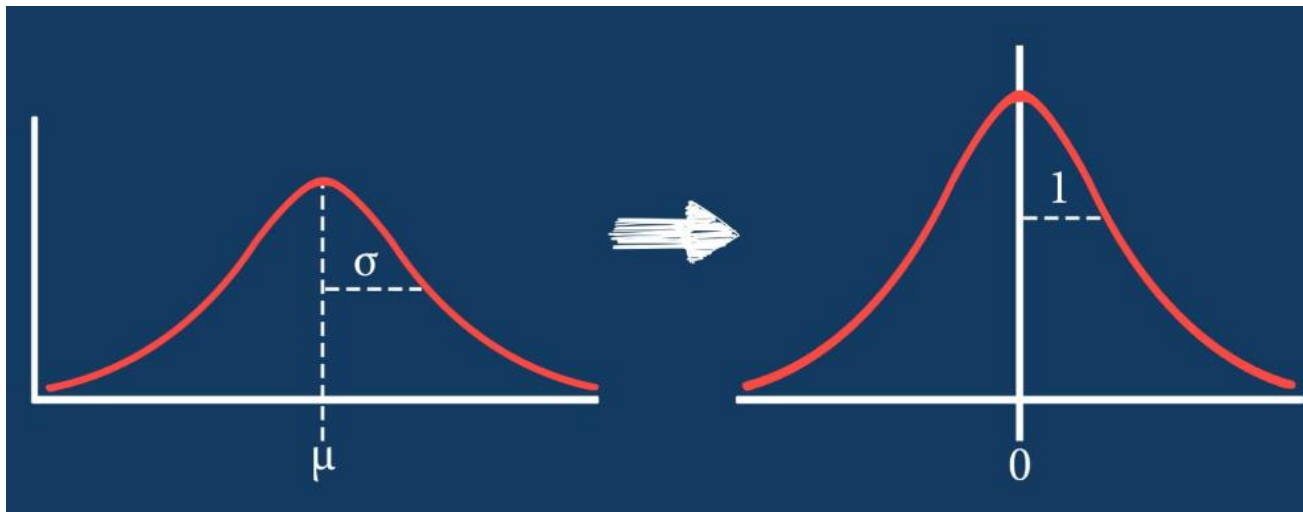
Example:

Suppose we have a dataset with two features: **Age** and **Income**.

- **Age** ranges from 20 to 80 years, while **Income** ranges from \$10,000 to \$100,000.
- If we don't standardize, **Income** might dominate the model because its range is much larger than **Age**.

After standardization:

- **Age** might be scaled to have a mean of 0 and a standard deviation of 1.
- **Income** will also be scaled similarly, making both features comparable in magnitude and ensuring that neither dominates the learning process.



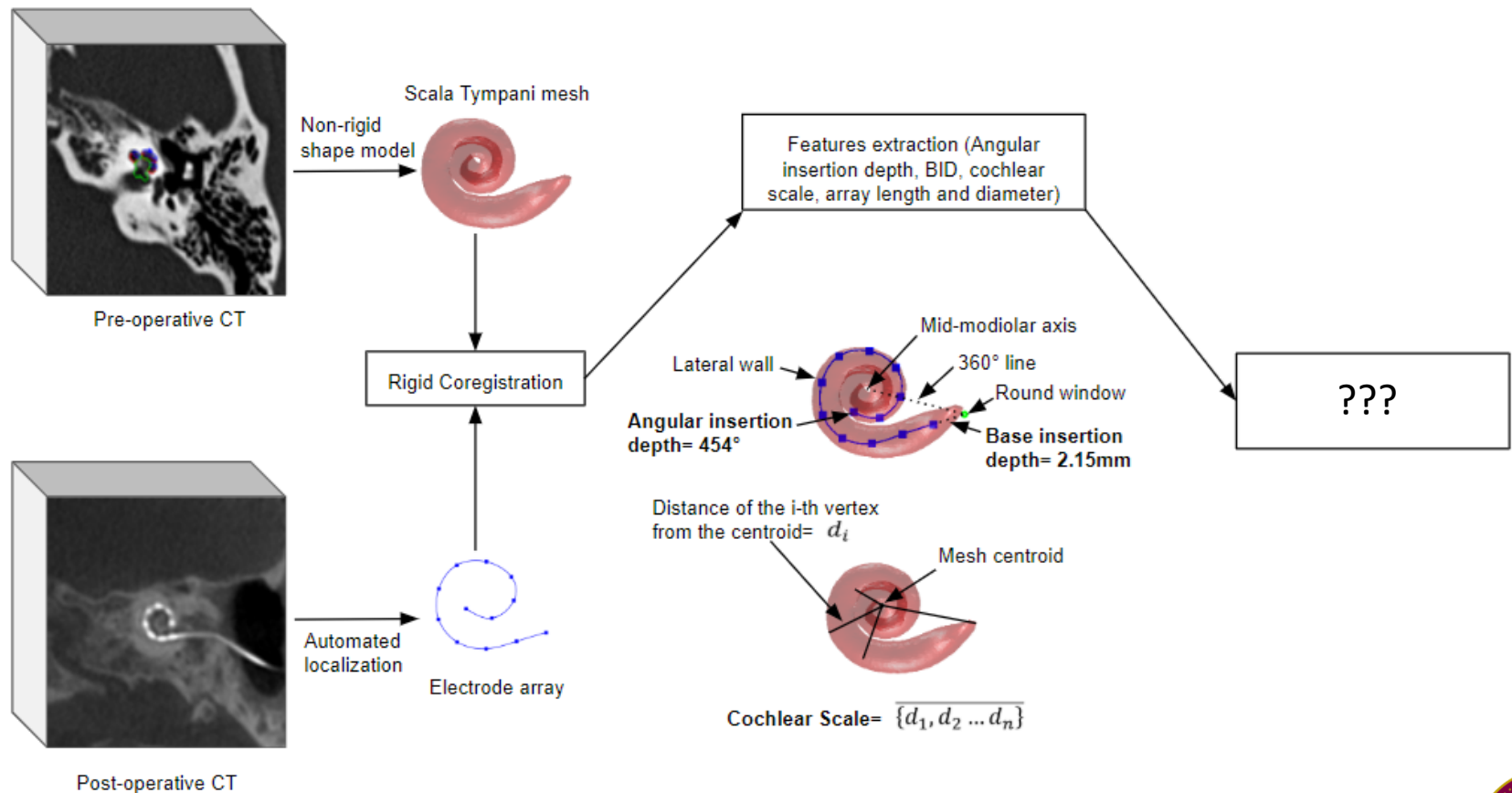
Random Forest Demo: Diabetes Classification

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.60000	0.62700	50	1
1	1	85	66	29	0	26.60000	0.35100	31	0
2	8	183	64	0	0	23.30000	0.67200	32	1
3	1	89	66	23	94	28.10000	0.16700	21	0
4	0	137	40	35	168	43.10000	2.28800	33	1
5	5	116	74	0	0	25.60000	0.20100	30	0
6	3	78	50	32	88	31.00000	0.24800	26	1
7	10	115	0	0	0	35.30000	0.13400	29	0
8	2	197	70	45	543	30.50000	0.15800	53	1
9	8	125	96	0	0	0.00000	0.23200	54	1
10	4	110	92	0	0	37.60000	0.19100	30	0
11	10	168	74	0	0	38.00000	0.53700	34	1
12	10	139	80	0	0	27.10000	1.44100	57	0
13	1	189	60	23	846	30.10000	0.39800	59	1
14	5	166	72	19	175	25.80000	0.58700	51	1
15	7	100	0	0	0	30.00000	0.48400	32	1
16	0	118	84	47	230	45.80000	0.55100	31	1
17	7	107	74	0	0	29.60000	0.25400	31	1
18	1	103	30	38	83	43.30000	0.18300	33	0
19	1	115	70	30	96	34.60000	0.52900	32	1

Random Forest Demo: BMI Prediction

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.60000	0.62700	50	1
1	1	85	66	29	0	26.60000	0.35100	31	0
2	8	183	64	0	0	23.30000	0.67200	32	1
3	1	89	66	23	94	28.10000	0.16700	21	0
4	0	137	40	35	168	43.10000	2.28800	33	1
5	5	116	74	0	0	25.60000	0.20100	30	0
6	3	78	50	32	88	31.00000	0.24800	26	1
7	10	115	0	0	0	35.30000	0.13400	29	0
8	2	197	70	45	543	30.50000	0.15800	53	1
9	8	125	96	0	0	0.00000	0.23200	54	1
10	4	110	92	0	0	37.60000	0.19100	30	0
11	10	168	74	0	0	38.00000	0.53700	34	1
12	10	139	80	0	0	27.10000	1.44100	57	0
13	1	189	60	23	846	30.10000	0.39800	59	1
14	5	166	72	19	175	25.80000	0.58700	51	1
15	7	100	0	0	0	30.00000	0.48400	32	1
16	0	118	84	47	230	45.80000	0.55100	31	1
17	7	107	74	0	0	29.60000	0.25400	31	1
18	1	103	30	38	83	43.30000	0.18300	33	0
19	1	115	70	30	96	34.60000	0.52900	32	1

Discussion on Assignment 1



Thank you